

Unit 09

Understanding the Internet of Things (IoT)

A. Warming Up

- What are the examples of Internet of Things that you can see or use in your lives?
스마트 냉장고, 스마트 워치, 스마트 밴드 등
- How does the Internet of Things work to improve our lives?
가전제품 인터넷에 연결되어 서로와 정보를 주고받아 편리하게 생활할 수 있음
- What can we expect in the future with the use of the Internet of Things?
의료, 농업, 공업, 교통 등 여러 분야에서 더 나은 삶을 기대함

B. Key Expressions

- Through this interconnected network, data can be transferred without the need for human-to-human or human-to-computer interaction.
- IoT relies on web-enabled smart devices equipped with embedded systems such as processors, sensors, and communication hardware.

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D. Comprehension Checkup

Choose the best answer to the following questions.

- What is the main characteristic of the Internet of Things (IoT)?
(A) Human-to-human interaction
(B) Human-to-computer interaction
(C) Data transfer without human interaction
(D) Data transfer without device interaction
- What kind of devices are essential for the functioning of IoT?
(A) Web-enabled smart devices
(B) Mainframe computers
(C) Non-internet-connected devices
(D) Virtual reality headsets
- How do IoT devices collect and transmit data for analysis?
(A) Through human intervention
(B) Locally stored in the devices
(C) Via IoT gateways or edge devices
(D) By using artificial intelligence algorithms

F. Vocabulary Checkup

Choose one that can best complete the following sentences.

- IoT automation plays a significant role in improving the quality of services and reducing the need for human _____.
(A) intervention (B) intention
(C) injection (D) convention
- Security is a major _____ as the number of connected devices increases.
(A) quality (B) concern
(C) tactic (D) discern
- Collecting, storing, and effectively utilizing data from numerous devices require advanced data management _____.
(A) states (B) status
(C) strategies (D) statistics
- Another potential _____ of IoT is the lack of an international standard for compatibility between devices from different manufacturers.
(A) advantage (B) benefit
(C) merit (D) drawback
- Addressing this challenge requires collaboration and the development of _____ standards and protocols to ensure seamless integration and interaction between IoT devices.

- Through this _____ network, data can be transferred without the need for human-to-human or human-to-computer interaction.
(A) interconnect (B) interconnected
(C) interconnecting (D) interconnection
- IoT relies on web-enabled smart devices _____ with embedded systems such as processors, sensors, and communication hardware.
(A) equip (B) equipping
(C) equipped (D) to equip
- The connectivity, networking, and communication protocols used in IoT vary _____ on the specific applications, and the integration of artificial intelligence (AI) and machine learning further enhances data collection processes.
(A) depend (B) depends
(C) depended (D) depending
- It is crucial to ensure robust security measures to protect sensitive data and prevent _____ access.
(A) authorized (B) authorizing
(C) unauthorized (D) unnaturalizing
- Improved communication between connected devices enables seamless data exchange, while _____ data packets over a connected network saves time and money.
(A) transfer (B) transferring
(C) transferred (D) to transfer

4. What is one of the major advantages of IoT?
- (A) Limited access to information
 - (B) Restricted data exchange
 - (C) Seamless data exchange from anywhere, at any time
 - (D) Costly and time-consuming data transfer
5. What is a significant challenge associated with IoT implementation?
- (A) Limited data collection capabilities
 - (B) Lack of security measures
 - (C) Insufficient processing power of devices
 - (D) Slow data transfer speeds
6. Why is interoperability between devices a challenge in IoT?
- (A) Lack of common standards and protocols
 - (B) Excessive data collection
 - (C) Incompatibility with web-enabled devices
 - (D) Insufficient networking capabilities

E. Grammar Checkup

Choose one that can best complete the following sentences.

1. With more devices interconnected, the potential for hackers to steal information grows.
- (A) confidential
 - (B) confident